

Final Year Design Project System Requirements Specification

AI Based Body Measurement App

Final Year Design Project SRS

by

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AI Based Body Measurement App

Executive Summary

This System Requirements Specification (SRS) defines the functional and non-functional requirements of the **AI-Based Body Measurement App**. The system extracts accurate body measurements from images using AI-based pose estimation models such as **ML Kit, MoveNet, or PoseNet**, with all processing performed **on-device** to ensure privacy and efficiency. It integrates **Firestore** for authentication and secure data storage and provides **personalized dress recommendations** based on extracted measurements. The application primarily supports **static image processing**, with optional support for **real-time camera input** depending on device capabilities.

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Requirements Analysis

This SRS outlines functional and non-functional requirements, system constraints, user roles, use cases, and interface specifications. The SRS defines functional and non-functional requirements, actors, use cases, interface requirements, constraints, system behavior, and traceability matrices.

User classes and characteristics

User Class	User Characteristics
End User	Uses the mobile application to upload or capture images, view extracted body measurements, receive personalized dress recommendations, and access measurement history. Requires an intuitive and privacy-preserving interface.
System Developer	Responsible for implementing application logic, integrating AI pose estimation models, and ensuring correct interaction between frontend and backend components.
Administrator	Manages the backend system, updates the dress catalog, maintains system configurations, and oversees user data stored in Firebase.
Maintenance Technician	Performs system updates, bug fixes, performance optimization, and ensures continuous system availability.
QA Specialist	Tests the system thoroughly to identify bugs, validate accuracy of measurements, and ensure performance meets requirements.
Security Expert	Evaluates and implements measures to protect sensitive user data such as body measurements and personal images.
Customer Support	Assists end users with troubleshooting, provides help using app features, and gathers feedback for improvements.

Requirement Identifying Technique

A systematic requirements identification process was adopted to ensure that all system requirements are complete, accurate, and aligned with project objectives. **Use Case Analysis** was selected as the primary technique, as the system is highly interactive and user-centric. Fully dressed use cases were developed to capture user–system interactions, define system behavior, and derive functional requirements in a structured manner. In addition, **storyboarding** was used to visually represent key user workflows such as image capture, measurement extraction, and personalized dress recommendations, ensuring a logical, clear, and user-friendly interaction flow. The combined use of these techniques ensures traceability between user needs, functional requirements, and system behavior.

Functional Requirements

The functional requirements of the system were derived primarily through **Use Case Analysis**, focusing on core functionalities such as **image input, body measurement extraction, secure data storage, and personalized dress recommendations**. Storyboarding further supported this process by visualizing key user journeys and validating the sequential and graphical flow of the application to enhance usability and consistency.

Functional Requirement 1: User Account Management

Table 1: Functional Requirements Specification for User Account Management

Identifier	FR-1
Title	User Registration and Authentication
Requirement	The End User shall be able to register and log in securely to access personalized features.
Source	Proposal Section 1.4 (Objectives - Enhanced User Experience, Privacy)
Rationale	To personalize experience, secure user data, and maintain a history of measurements.
Business Rule	User email must be unique. Passwords must be at least 8 characters long
Dependencies	None
Priority	High

Functional Requirement 2: Image Input Functionality

Table 2: Functional Requirements Specification for Image Input

Identifier	FR-2
Title	Image Input Functionality
Requirement	The End User shall be able to upload or capture images using the device camera or upload an image from gallery.
Source	Project Requirement Analysis
Rationale	Users must be able to provide up-to-date and accurate images for measurement.
Business Rule	Images must meet minimum resolution and clarity requirements defined by the model.
Dependencies	None
Priority	High

Functional Requirement 3: *Body Keypoint Detection*

Table 3: Functional Requirements Specification for Body keypoint Detection

Identifier	FR-3
Title	<i>Body Keypoint Detection</i>
Requirement	The system shall process the image locally using integrated MoveNet/ML Kit model to detect 17 body keypoints.
Source	Project Requirement Analysis
Rationale	Keypoint detection is essential for calculating body measurements.
Business Rule	No image shall be sent to external servers for processing.
Dependencies	FR-2
Priority	High

Functional Requirement 4: Measurement Calculation

Table 4: Functional Requirements Specification for Body Measurement Extraction

Identifier	FR-4
Title	Body Measurement Calculation
Requirement	The system shall compute accurate body measurements (e.g., chest, waist, hip, arm length) from detected keypoints.
Source	system architecture description
Rationale	Computation of measurements is the core objective of the application.
Business Rule	Measurement accuracy shall remain within the tolerance defined by selected pose estimation model
Dependencies	FR-3
Priority	High

Functional Requirement 5: Display Measurement

Table 5: Functional Requirements Specification for Display Body Measurement

Identifier	FR-5
Title	Display Body Measurements
Requirement	The system shall display detected keypoints and measurement values.
Source	User Interface Requirements
Rationale	Representation helps users understand measurement accuracy.
Business Rule	Keypoints must be displayed clearly and must not overlap with critical UI elements.
Dependencies	FR-4
Priority	Medium

Functional Requirement 6: Measurement Storage

Table 6: Functional Requirements Specification for Body Measurement Storage

Identifier	FR-6
Title	Measurement Storage
Requirement	The system shall store user measurements securely for future access.
Source	User needs
Rationale	Users need historical records to track progress or reuse measurements for recommendations.
Business Rule	Data must be stored in an encrypted, accessible format.
Dependencies	FR-4
Priority	Medium

Functional Requirement 7: Dress Catalog & Personalized Recommendation

Table 7: Functional Requirements Specification for Dress Recommendation

Identifier	FR-7
Title	Browse and Get Recommended Dresses
Requirement	The End User shall be able to browse a catalog of pre-made dress designs. The system shall recommend designs and suggest sizes based on the user's most recent body measurements .
Source	Proposal Section 1.4 (Objectives - Integration with Retail Platforms)
Rationale	To translate body measurements into tangible value, enabling informed fashion choices and reducing online shopping returns.
Business Rule	Recommendations are generated using a predefined measurement-to-size mapping algorithm.
Dependencies	FR-6,FR-4
Priority	High

Non-Functional Requirements

Reliability

The system shall achieve at least **90% measurement accuracy** under standard lighting conditions. The application shall have a **Mean Time Between Failures (MTBF) of at least 30 days** and shall recover from application crashes within **90 seconds** without data loss.

Usability

Users shall be able to complete the full body measurement process within **a maximum of five user interactions**. The application interface shall be easy to learn, requiring no more than **10 minutes of initial usage** for a new user to perform core tasks successfully.

Performance

The system shall generate body measurement results within **90 seconds** after image submission. User interface response time for navigation actions shall not exceed **2 seconds** under normal operating conditions.

Security

All sensitive user data, including images and body measurements, shall be processed on-device or stored in Firebase using **end-to-end encryption**. Unauthorized access attempts shall be logged, and user data shall only be accessible to authenticated users.

External Interface Requirements

This section defines how the system interacts with users and external software or hardware components to ensure smooth and reliable communication.

User Interface Requirements

The application shall follow **Google Material Design guidelines** to ensure consistency and usability. The user interface shall support readable font sizes, appropriate color contrast for accessibility, and intuitive navigation through icons and buttons. Screens include image preview, measurement results, recommendation lists, and history views.

Software Interfaces

The system shall integrate with **Firestore or Realtime Database** for data storage, **ML Kit Pose Detection API**, and the **TensorFlow Lite interpreter** for on-device inference.

Hardware Interfaces

The system requires a smartphone equipped with a camera, sufficient memory, and processing capability to support on-device AI inference. Optional sensors such as a gyroscope may be used for image orientation handling.

Communication Interfaces

An internet connection is required only for Firebase-related operations such as authentication and data synchronization. The body measurement and pose detection modules shall function **offline**.

Use case Analysis

UC1 – Extract Body Measurements

Table 8: Use case Description for Extract Body Measurements

Field	Description
UC Identifier	UC1
Requirements Traceability	FR-2, FR-3, FR-4, FR-6
Purpose	Extract accurate body measurements from user images using AI pose detection.
Priority	High
Preconditions	User is logged in and camera permission is granted.
Postconditions	Body measurements are calculated, displayed, and stored.
Actors	End User
Extends	None
Main Success Scenario	1. User uploads or captures an image. 2. System detects body keypoints. 3. Measurements are calculated. 4. Results are displayed to the user.
Alternate Flows	Image quality is insufficient; system prompts user to retry.
Exceptions	Camera failure or model loading error.
Includes	UC2

UC2 – Upload Image

Table 9: Use case Description for Upload Image

Field	Description
UC Identifier	UC2
Requirements Traceability	FR-2
Purpose	Allow user to capture or upload an image.
Priority	High
Preconditions	Camera or gallery access granted.
Postconditions	Image is temporarily stored for processing.
Actors	End User
Extends	None
Main Success Scenario	1. User selects upload option. 2. Camera or gallery opens. 3. User confirms image selection.
Alternate Flows	User cancels the upload.
Exceptions	Permission denied.
Includes	None

UC3 – View Dress Recommendations

Table 10: Use case Description for Dress Recommendation

Field	Description
UC Identifier	UC3
Requirements Traceability	FR-7
Purpose	Provide personalized dress recommendations and size suggestions based on the user's body measurements.
Priority	Medium
Preconditions	User is logged in and body measurements are available.
Postconditions	Recommended dress list is displayed to the user.
Actors	End User
Extends	None
Main Success Scenario	1. User selects the "View Recommendations" option. 2. System retrieves the latest body measurements. 3. System matches measurements with available dress sizes. 4. Recommended dresses and sizes are displayed.
Alternate Flows	No suitable dress is found; system displays a message indicating no matches.
Exceptions	Database retrieval failure.

Includes	UC1
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UC4 – Save Measurement History

Table 11: Use case Description for Save Measurement History

Field	Description
UC Identifier	UC4
Requirements Traceability	FR-6
Purpose	Save extracted body measurements for future reference and reuse.
Priority	Medium
Preconditions	User is logged in and measurements have been generated.
Postconditions	Measurement record is stored securely in the database.
Actors	End User
Extends	None
Main Success Scenario	1. User selects the “Save Measurements” option. 2. System validates measurement data. 3. System stores the data securely.
Alternate Flows	User cancels the save operation.
Exceptions	Database connectivity failure.
Includes	UC1

UC5 – View History

Table 12: Use case Description for View History

Field	Description
UC Identifier	UC5
Requirements Traceability	FR-6
Purpose	Allow the user to view previously saved body measurements.
Priority	Low
Preconditions	User is logged in and measurement history exists.
Postconditions	Measurement history is displayed to the user.
Actors	End User
Extends	None
Main Success Scenario	1. User selects the “View History” option. 2. System retrieves stored measurement records. 3. Measurement history is displayed.

Alternate Flows	No history exists; system displays an informational message.
Exceptions	Network or database access error.
Includes	None

UC6 – Admin Updates Dress Catalog

Table 13: Use case Description for Admin Updates catalogue

Field	Description
UC Identifier	UC6
Requirements Traceability	FR-7
Purpose	Allow the administrator to add, update, or remove dress designs in the catalog.
Priority	Low
Preconditions	Administrator is authenticated.
Postconditions	Dress catalog is updated successfully.
Actors	Administrator
Extends	None
Main Success Scenario	1. Administrator logs into the system. 2. Administrator selects catalog management option. 3. Administrator adds, edits, or deletes a dress entry. 4. System saves the changes.
Alternate Flows	Invalid data entered; system prompts correction.
Exceptions	Permission violation or database failure.
Includes	None

UC7 – Notifications and Alerts

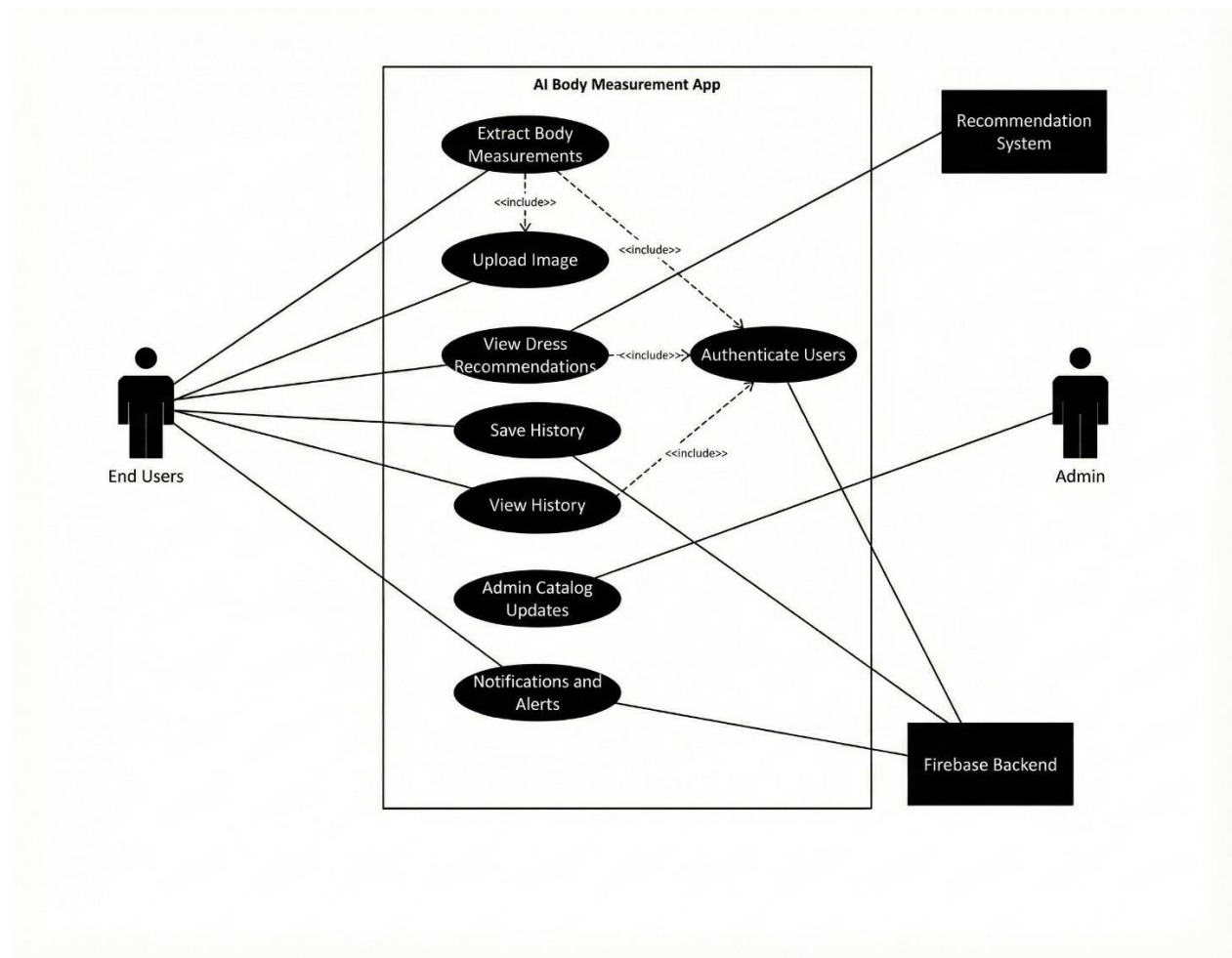
Table 14: Use case Description for Notification and Alerts

Field	Description
UC Identifier	UC7
Requirements Traceability	FR-1
Purpose	Notify users about important events such as measurement completion or system updates.
Priority	Low
Preconditions	User has enabled notifications.
Postconditions	Notification is delivered to the user.

Actors	End User
Extends	None
Main Success Scenario	1. A system event occurs. 2. System generates a notification. 3. User receives the alert.
Alternate Flows	Notifications are disabled by the user.
Exceptions	Notification delivery failure.
Includes	None

Use Case Diagram

The use case diagram illustrates the interactions between external actors and the AI-Based Body Measurement App. Actors include the End User and Administrator, who interact with use cases such as Upload Image, Extract Body Measurements, View Recommendations, Save Measurement History, and Update Dress Catalog. The diagram clearly defines the system boundary and ensures that all user interactions are traceable to functional requirements.



Storyboards

S-1 Home



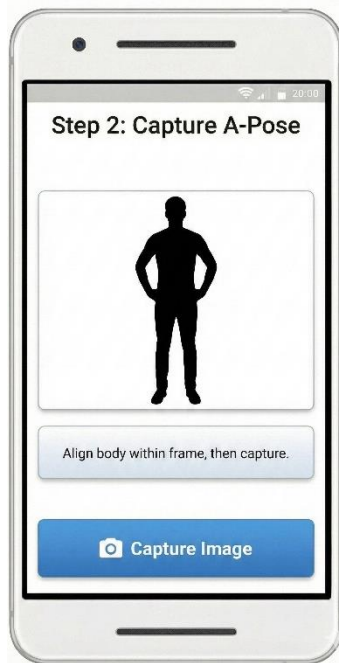
User views main dashboard with options to capture image, view history, or browse recommendations.

S-2 Guidance Screen



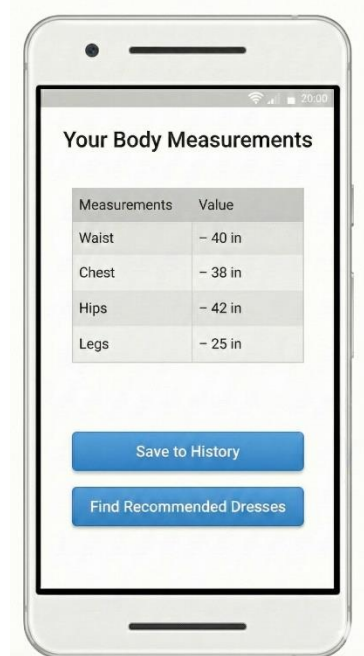
Application provides instructions for correct posture and image capture.

S-3 Capture Image



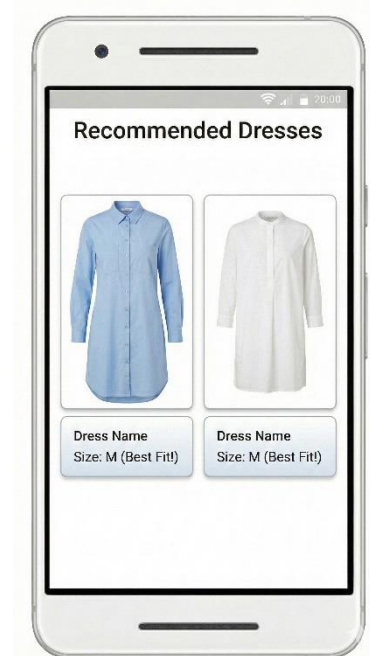
User captures or uploads an image.

S-4 Display results



Extracted body measurements are shown visually.

S-5 Recommended Dresses



Personalized dress suggestions are displayed based on measurements.

Summary

This System Requirements Specification (SRS) presents a comprehensive and structured definition of the requirements for the **AI-Based Body Measurement App**. The document establishes a clear understanding of the system's purpose, scope, functional behavior, non-functional constraints, external interfaces, and user interactions. The primary objective of the system is to accurately extract human body measurements from user images using AI-based pose estimation techniques while ensuring privacy, usability, and reliability.

A systematic **requirements engineering process** was followed to ensure that the system requirements are complete, consistent, verifiable, and aligned with project goals. The requirements identification phase employed **use case analysis and storyboarding techniques** to capture user expectations and system behavior effectively. These techniques helped translate high-level project objectives such as privacy-preserving measurement extraction, personalized dress recommendations, and improved user experience into well-defined functional and non-functional requirements.

During requirements identification, stakeholders and user classes were analyzed to understand their roles and expectations. End users, administrators, developers, and maintenance personnel were considered to ensure that both operational and technical needs were addressed. Functional requirements were derived by identifying core system operations, including user authentication, image input, body keypoint detection, measurement calculation, storage, and personalized recommendations. Non-functional requirements were defined to enforce system quality attributes such as performance, usability, reliability, and security, which are critical for user trust and system adoption.

Use case analysis played a central role in refining system functionality. Fully dressed use cases were developed to describe user-system interactions in a structured manner, specifying actors, preconditions, postconditions, main success scenarios, alternate flows, and exceptions. This approach ensured that all system behaviors were traceable to functional requirements and supported realistic usage scenarios. Use cases such as "Extract Body Measurements," "Upload Image," "View Dress Recommendations," and "Save Measurement History" provided a clear operational view of the system and guided interface design and implementation planning.

The integration of requirements engineering practices ensured that the system design remains aligned with the project's objectives. By clearly defining requirements early in the development lifecycle, the SRS minimizes ambiguity, reduces development risks, and provides a reliable reference for implementation, testing, and validation. Furthermore, traceability between objectives, requirements, and use cases ensures that every system feature delivers measurable value to users.

In conclusion, this SRS serves as a foundational document that bridges stakeholder expectations and technical implementation. The structured requirements identification process and thorough use case analysis ensure that the AI-Based Body Measurement App is scalable, secure, user-centric, and capable of achieving its intended goals of accurate body measurement extraction, privacy preservation, and enhanced decision-making in fashion selection.

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